LAST30DAYSYT · TOPIC BRIEF

AI local models, last 30 days: the year running AI on your own machine stopped being a hobby

An open model overtook the closed frontier, Apple shipped a first-party local-agent stack, and the best-watched videos all answered the same question — how do I run capable AI without the cloud or the bill?

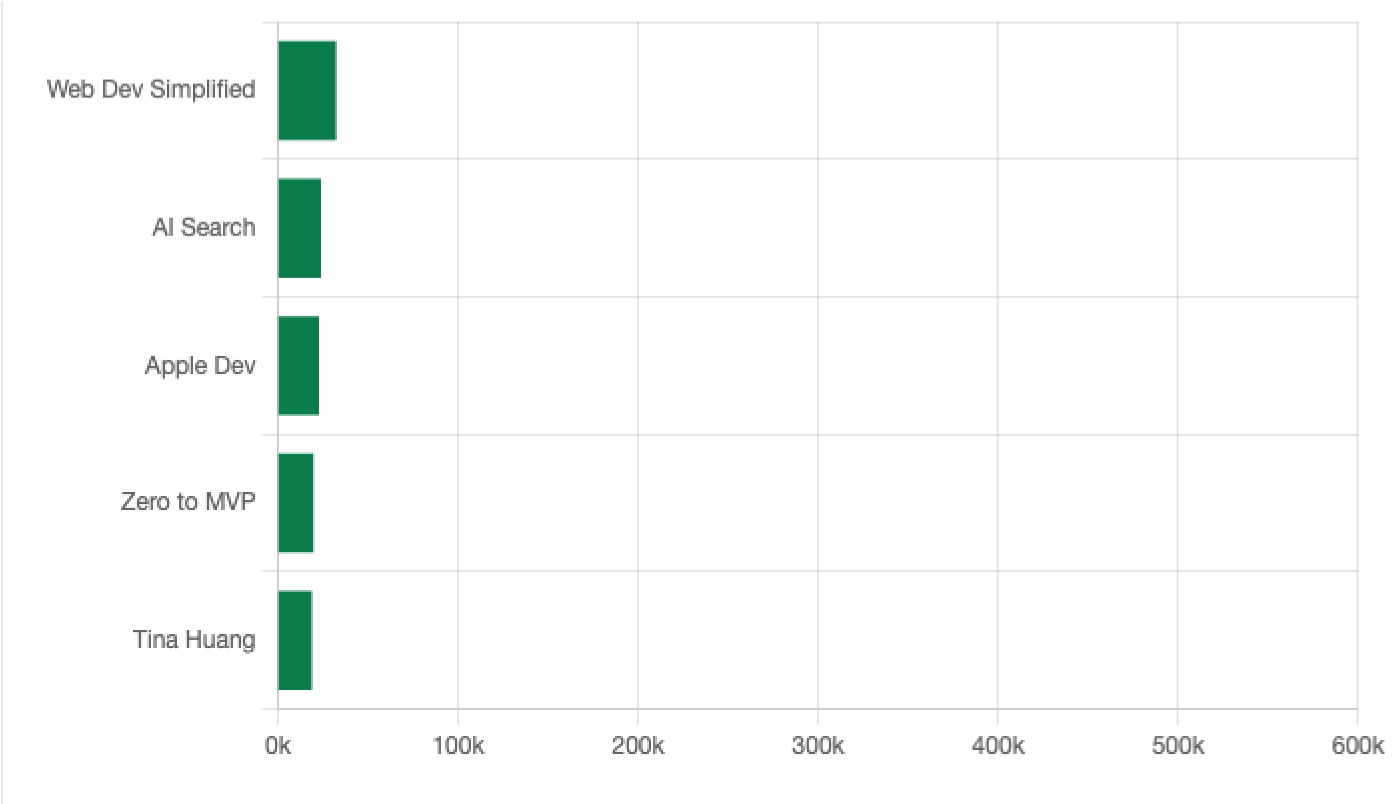
Topic: "AI local models" · 5 top videos · window 2026-03-24 → 2026-06-22 · velocity-ranked (views/day), Shorts & live excluded · generated by last30daysYT

Each claim is tagged for how far we could verify it beyond the video: CONFIRMED corroborated by an outside source · SINGLE SOURCE only the video says it · OVERSTATED the video oversells what the evidence shows.

TL;DR

- **The best open model closed most of the gap — but didn't "beat everything."** OVERSTATED The video says GLM 5.2 beats the best GPT and Gemini; independent benchmarks show it *wins* SWE-bench Pro yet *trails* on Terminal-Bench — at ~1/6 the cost. [00:00] · VentureBeat
- **Apple made local agents first-class.** CONFIRMED A WWDC26 session runs a full agentic loop on-device via MLX — "no cloud, no API keys." [00:07] · Apple Developer
- **Local agents are "a new category."** SINGLE SOURCE One primer's framing: agent = LLM brain + text-file memory + tools + a cron "heartbeat." [00:00]
- **The motivation is money.** SINGLE SOURCE The most-watched video (552k) opens flatly: "AI pricing is outrageous." [00:00]
- **It runs on a Raspberry Pi now.** CONFIRMED Gemma 4 E2B answers real prompts on an 8GB Pi 5 at 3–8 tokens/sec — slow, but usable for scripts. [08:48] · independent test

ENGAGEMENT — VIEWS PER FEATURED VIDEO



THEME 1 — THE MODELS GOT GOOD ENOUGH

An open model closed most of the gap OVERSTATED

The single biggest story is ZAI's **GLM 5.2**. The reviewer's claim is blunt: "We have a new number one open-source model and the gap is insane... it even beats the best GPT and the best Gemini across multiple benchmarks." [00:00] The demos are real — in single prompts it built an interactive 3D digital twin of Earth, an exploded-view V8 engine, and a mechanical watch with turning gears.

But the blanket "beats everything" doesn't survive the benchmarks. Independent numbers show a split decision: GLM 5.2 *wins* SWE-bench Pro (62.1% vs GPT-5.5's 58.6% and Gemini 3.1 Pro's 54.2%) but *trails* on Terminal-Bench 2.1 (81.0 vs GPT-5.5's 84.0 and Claude Opus 4.8's 85.0). Where it's unambiguous is **cost** — roughly one-sixth of the closed frontier — and open HTML design, where it topped Design Arena on 2026-06-19. EdenAI benchmarks · VentureBeat

> *"The thing I like about GLM is it requires very minimal handholding. It just works right out of the box most of the time." [06:47]*

Why it matters: the honest read isn't "open won" — it's "open now competes head-to-head on some tasks at a fraction of the price." For a weight you can download, that's the real story; the "beats everything" framing is the hype tax.

...and it now fits on an $80 computer CONFIRMED

At the other end of the scale, a developer runs **Gemma 4**'s smallest model (E2B, ~4B params) on a Raspberry Pi 5 with 8 GB of RAM. [01:03] It's Apache-2.0 licensed with a 128k context and native audio/image support. [02:37] Independent tests line up with the video: at Q4 it needs ~2–3 GB and returns 3–8 tokens/sec via llama.cpp — slow, but usable for scripts and automation, not real-time chat. Pi 5 benchmark

> *"It's absolutely possible, and the results are usable, at least for certain scripts or automation tasks." [08:48]*

WHAT ACTUALLY RUNS LOCALLY — AND HOW FAST

Setup	Model	Quant	Needs	Speed	Verdict
Raspberry Pi 5 (8 GB)	Gemma 4 E2B (~4B)	Q4_K_M	~2–3 GB	3–8 tok/s	Usable for scripts/automation
Raspberry Pi 5 (8 GB)	Gemma 4 E4B (~4B)	Q4_K_M	~3–4 GB	2–4 tok/s	Works, noticeably slower
Any GPU	rule of thumb	Q4 ≈ ½ of Q8	model must fit VRAM; overflow spills to system RAM → slow	—	Start at Q4, size up if VRAM allows

Why it matters: "local" no longer requires a GPU rig — the floor is a board you can hide behind a monitor. The table is the decision: pick the variant and quant your memory budget allows, and expect tokens-per-second, not chat latency. (Sources: DEV.to test, benchmark.)

THEME 2 — THE AGENT LAYER MATURED

Apple shipped a first-party local-agent stack CONFIRMED

The most consequential release this month is institutional: a WWDC26 session walks through running the full **agentic loop** — user → agent → model → tools, cycling until the task is done — entirely on a Mac via MLX. [01:10] It's a clean four-layer stack: MLX → MLX-LM → an OpenAI-compatible MLX-LM Server → any agent (Xcode, OpenCode, Pi). [02:46]

> *"No cloud, no API keys, just your hardware doing the work." [00:07]*

The hardware angle is concrete: the M5's Neural Accelerators make matrix multiply ~4× faster than M4, which maps almost directly to prompt-processing speed [06:25] — the bottleneck for agents, which churn hundreds of thousands of tokens of tool output. For models too big for one machine (DeepSeek's 1.6T params need >800 GB), MLX shards across

Macs over Thunderbolt. [07:58] Why it matters: when the platform vendor ships the stack, local agents move from enthusiast project to supported workflow. (Apple Developer, WWDC26 session 232)

"Local AI agents are a new category of AI product" SINGLE SOURCE

The most useful mental model came from a 26-minute primer. Its anatomy of an agent is refreshingly demystifying: pick where it lives, give it a communication channel (Telegram/Discord), a brain (the LLM), memory, tools, and a "heartbeat" (a cron schedule so it acts unprompted). [02:35]

> *"Memory is literally just a bunch of text files... you just dump in there everything that you want it to know." [06:44]*

The honest part is the safety warning: people who gave an agent access to everything on their main laptop had it delete emails — so she runs hers on a wiped, dedicated old MacBook. [03:36] Why it matters: the gap between "I have a local model" and "I have a useful local agent" is design, and this is the clearest map of it.

THEME 3 — THE REALITY CHECK: HARDWARE & COST

The most-watched video is a VRAM lesson, and that's the tell CONFIRMED

552k people watched a 45-minute local-coding setup whose real subject is hardware literacy. The opener names the motive — "AI pricing is outrageous" [00:00] — then teaches the three numbers that decide everything: parameters, context, and VRAM. The key constraint: a model loads into your GPU's VRAM, and whatever doesn't fit overflows to system RAM and slows down. [04:07]

> *"The key takeaway: how much RAM your graphics card has, as well as how much RAM your computer overall has." [04:38]*

It also demystifies **quantization** (Q4/Q6/Q8) — rounding the weights to shrink the model, with Q4 a good starting point and roughly half the size of Q8. [07:45] Why it matters: the demand signal is loud — the single biggest audience this month went to "how do I actually fit this on my machine and stop paying per token."

WORTH WATCHING

- New #1 open-source AI model is here! — AI Search · 407k views. The GLM 5.2 demos; start here for "how good is open now."

- WWDC26: Run local agentic AI on the Mac using MLX — Apple Developer · 389k views. The most authoritative how-it-works, 13 min.
- The Best Local Agentic Coding Workflow — Web Dev Simplified · 552k views. The deepest hands-on; the VRAM/quantization masterclass.

ON THE RADAR

Three threads to track next month: **GLM 5.2's benchmark picture is now in — and it's mixed** (wins SWE-bench Pro, trails Terminal-Bench), so watch whether the next point release closes the agentic-terminal gap or just the price gap; **whether M5 / MLX makes on-device agents fast enough to replace cloud coding day-to-day**; and **the agent-safety story** — the "it deleted my emails" warning suggests sandboxing, not capability, is the next bottleneck.

Built by **last30daysYT** from YouTube transcripts (via yt2md). Topic "AI local models", top 5 by velocity (views/day), last 30 days (extended to a 90-day window to fill the set). Every claim links to the exact moment in the source video; headline claims were then checked against an outside source and tagged CONFIRMED / SINGLE SOURCE / OVERSTATED. Engagement is a prefilter — substance was judged from the transcripts and the corroboration pass.